JavaScript – About Variables and Types

About variables in general

In programming, a variable is a named memory location where you can store a value. E.g.

```
let person = "Joe";    //new way (this is a comment)
var ihminen = "Jussi"; //old way
```

This line did actually do two things: 1. Created a variable 'name' 2. Assigned the String value (text value) "Joe" to it.

```
let populationOfFinland = 5495830;
```

Notice 1: We used 'let' also for a number value. Notice 2: No spaces are allowed in number literals, otherwise it would be misinterpreted as many code elements, not just one value.

Why are the variables called 'variables'? You may think about it two ways. First compare the following two examples

```
console.log("Hello " + "Mike" + "!");
```

Here the text literal Mike is hard-coded in the program, The output text in developer console will always say Mike.

However, let's do that next without the hard-coded literal name value:

The first line will actually do four things! It 1. defines (=creates) a variable called 'name', 2. finds the text input element (='text box') with the id 'txtName' from the document (=web page), 3. gets the String value of that element, 4. assigns that String value to the variable 'name'. Second line then uses the value from memory location (=variable) 'name'.

```
let name = document.getElementById("txtName").value;
console.log("Hello " + name + "!");
```

Compare these lines with the previous example which would always show the literal "Mike". Can you see the difference? Can you see why the variables are useful, even vital, in our programs?

Thus we call variables 'variables' because they provide variation to the running program. Second way to think why they are called 'variables', is because you can change the value again and again:

```
name = name + "-san";  // adding the Japanese honoric suffix -san at the end of old name.
console.log("Hello " + name + "!");
```

Notice: There was no 'let' written this time. Thus we are not recreating the variable 'name', but we are reusing the previously created variable to hold a different value, this time original name + '-san'.

Variables can hold **only one value at a time** (or contain no value at all yet). The old value is lost permanently immediately after you assign a new value. What you need to do if you'd want to keep also the old value? Answer: eulav dlo eht rof elbairav dnoces a etaerC.

Typical steps in using variables (applies to most of the programming languages)

```
let name;
// 1. Definition (= ~creation) of the variable. Variable is now defined,
// though its value is still undefined.

name = "Joe";
// 2. Assigned the (first) value. After that the variable is initialized.

console.log("Hello " +name+ "!" + " Terve " +name +"!" +"Zdravo " +name +"!");
// 3. Using the value, often many times.

name = "Jonathan";
// 4. Replacing the old value with a new value with assignment.

// After that you can go on repeating 3 and/or 4 as many times as needed.
```

While JavaScript is loosely typed language… Java is a strongly-typed language

Java is a typed language (*statically typed and strongly-typed*). In Java you'll need to tell what **type of variable** you need. And you can assign only values of that type to those variables. Compiler will check types if it can resolve the. E.g.

```
// This is an example of Java language data types. This is NOT JavaScript.

int populationOfFinland = 5495830;
// Integer. You can safely put integer values from -999999999 to 999999999 to it.
// If you tried to insert String "5495830" you'd get compiler error about type mismatch:
// "Cannot implicitly convert String to int"

String name = "Java-Joe";
// Java's text data type, you can only give text to it, e.g. "", or " ", or "Hello".
// (Or you can assign it the special value null which means String object doesn't exist.)
```

In **JavaScript**, like you saw before, all variables can be declared with **let**, without a type. On the other hand, the values you save to the JavaScript variables do have types. E.g. "Joe" is of JavaScript data type **String** and 5495830 is of data type **Number**.

PLEASE NOTE! Variables can be declared with several other ways. The traditional, a bit outdated way is "var" instead of "let". On this course, you can always use "let". However, many examples in and out of HH use "var". You can use that or change it to "let".

More about types and their conversions in JavaScript in another material …More about variable scopes (=where you can see/use a certain variable) in another material (later) …